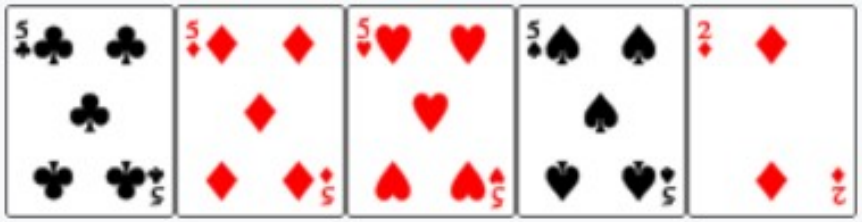
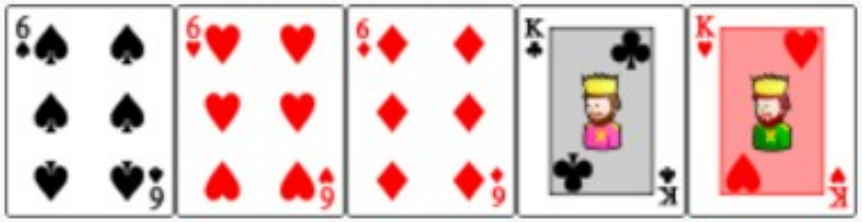
Poker

Exercises 4

Four of a kind	
Full house	

- How many hands contain 4-of a kind?
- Number of ways to select the kind of the 4-of a kind C(13,1).
- Number of ways to select the 5th card: 48 (12 for the rank times 4 for the suit).
- Total number is: $13 \cdot 48$
- How many hands contain a full house?
- Number of ways to select the 2 kinds C(13,2).
- Number of ways to select the kind for the 3-of a kind C(2,1).
- Number of ways to select the suits for the 3-of a kind C(4,3)
- Number of ways to select the suits for the pair C(4,2).
- Total number is: $C(13,2) \cdot C(2,1) \cdot C(4,3) \cdot C(2,2)$
- Poker Hands: Royal Flush > Straight Flush > 4-of a kind > Full house > Flush > Straight > 3-of a kind > 2 pairs > 1 pair